

Physics-Informed PMU Event Detection

and Localization

ExtraTrees-based pipeline for PMU event detection and localization

ExtraTrees

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SGSMA 2026 Challenge

Pipeline

1. Convert raw PMU waveforms into physics-aware features.
2. Use a hierarchical ExtraTrees classifier to identify event type.
3. Route each event to a specialized BUS, LINE, or PMU localizer.
4. Produce the challenge CSV output at timestamp resolution.

Reimplementation target

The method can be rebuilt from PMU windows, feature definitions, ExtraTrees models, and typed localization routes.

Submission boundary

The submitted route is SGMS: physics-informed features, hierarchical ExtraTrees, and typed dynamic localizers. RAW-label-tuned and oracle branches are excluded from inference.

Input

Multiple PMU CSV files, usually one per observed bus:

- ▶ synchronized timestamps,
- ▶ voltage and current phasors,
- ▶ frequency and ROCOF,
- ▶ data-quality indicators.

Output

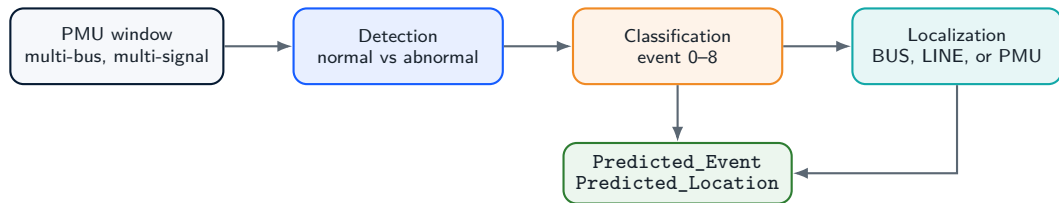
The submission must generate exactly:

- ▶ `TIMESTAMP`
- ▶ `Bus`
- ▶ `Predicted_Event`
- ▶ `Predicted_Location`

Scoring target

Detection is not enough. The system must also classify the event type and localize its origin or the failed PMU.

The task is three coupled decisions



Design consequence

The pipeline keeps these decisions separate: a physical classifier, missing/bad-data detectors, and typed localizers replace one flat classifier.

Label	Name	Physical interpretation
0	normal	No physical disturbance and no major data issue.
1	fault	Strong voltage disturbance, fast derivative, sag or transient.
2	line outage	Current redistribution without fault-level voltage collapse.
3	generation change	Frequency/ROCOF and active-power proxy change.
4	load change	Sustained current and power-consumption change.
5	missing data	Data loss without strong physical evidence.
6	missing + physical	Data loss combined with a real physical event.
7	bad data	One sensor/channel becomes inconsistent with the rest.
8	mixed proxy	Mixed physical signature not cleanly explained by 1–4.

Electrical measurements

Group	Columns
Voltage magnitude	VA_MAG, VB_MAG, VC_MAG
Current magnitude	IA_MAG, IB_MAG, IC_MAG
Voltage angle	VA_ANG, VB_ANG, VC_ANG
Current angle	IA_ANG, IB_ANG, IC_ANG
Dynamics	Freq, ROCOF

Support variables

- ▶ **TIMESTAMP**: time alignment.
- ▶ **DATA_PRESENT**: measurement availability.
- ▶ Bus ID from `Bus*.csv`.
- ▶ IEEE 39-bus topology for graph features.

Interpretation

Every feature family is built from a physical question: how large, how fast, how persistent, how coherent, and where in the grid?

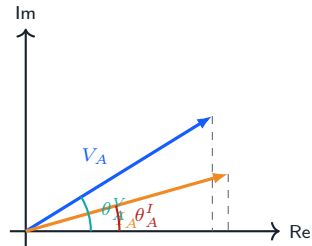
Per bus, per phase

$$V_{\phi}(t) = |V_{\phi}(t)|e^{j\theta_{\phi}^V(t)}, \quad I_{\phi}(t) = |I_{\phi}(t)|e^{j\theta_{\phi}^I(t)}$$

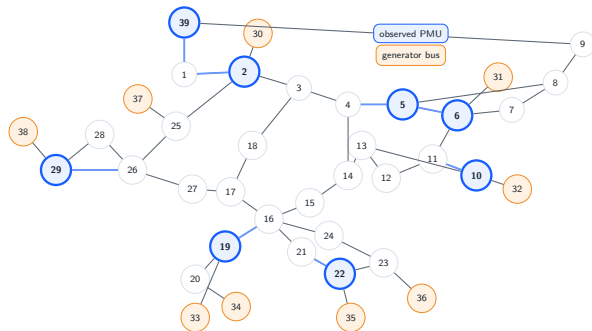
$$\phi \in \{A, B, C\}, \quad \dot{f}(t) = \text{ROCOF}(t)$$

Feature inputs

The extractor uses magnitudes, unwrapped angles, frequency, ROCOF, and data availability. It does not solve a full AC power-flow at inference time.



Power proxy comes from the angle gap $\theta^V - \theta^I$ and normalized magnitudes.



Observed PMUs

{2, 5, 6, 10, 19, 22, 29, 39}

Electrical reading

The localizer sees eight sparse PMU views. Events must be inferred from the disturbance pattern across those locations.

Topology used

Electrical distances come from the IEEE 39-bus admittance model, not drawing geometry.

Admittance and impedance

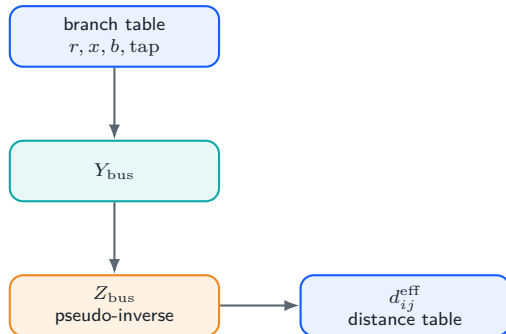
Y_{bus} : network admittance matrix

$$Z_{\text{bus}} = Y_{\text{bus}}^+$$

$$d_{ij}^{\text{eff}} = |Z_{ii} + Z_{jj} - Z_{ij} - Z_{ji}|$$

Meaning

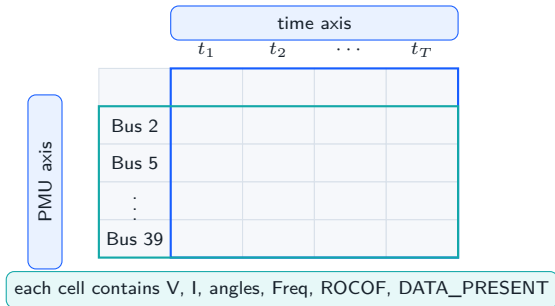
Two buses are close if an injection at one bus is expected to have a similar voltage effect at the other.



Where it appears

Graph-temporal features weight PMU differences by $1/d_{ij}^{\text{eff}}$.

Case representation before feature extraction



The extractor reads a tensor

Think of one case as a table over PMUs and time. Each table cell has several electrical channels.

Extracted representation

The window becomes one row of engineered features. Models never see the raw tensor directly.

Entrypoint

```
python main.py -input-dir <folder> -output  
<csv>
```

Active route

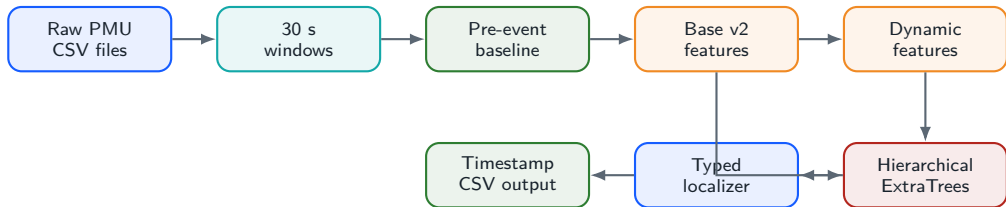
- ▶ `model = sgms_extra_trees_windowed_v2`
- ▶ `route = ml_windowed`
- ▶ `window_seconds = 30`

Call chain

```
main.py  
→ src/models/hybrid_submission.py  
→ src/data_factory/final_model.py
```

Fallback

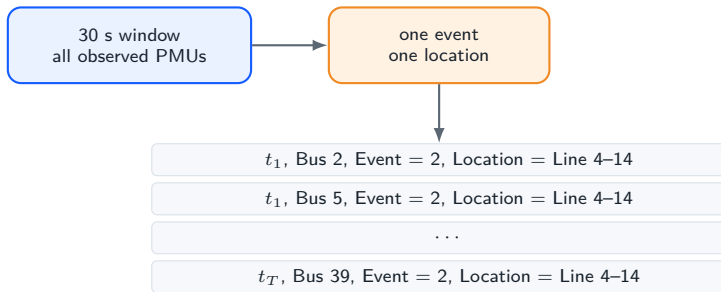
A bus-agnostic physics runtime remains only as a missing-bundle fallback.



Runtime shape

We convert each PMU window into a physics-aware signature, classify the event with guarded ExtraTrees models, then localize it with event-specific localizers.

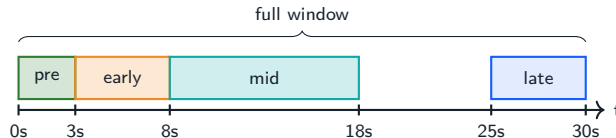
Window predictions to timestamp rows



Submission contract

The official CSV is timestamp-aligned even though the decision is made at the window level.

30-second window design



Window evidence

- ▶ One sample is too noisy.
- ▶ A fault has onset and transient.
- ▶ Load/generation changes are often slower.
- ▶ Missing data needs run-length evidence.

Runtime behavior

The model predicts once per window and replicates the window prediction to all required timestamp/bus rows.

Given one 30-second window

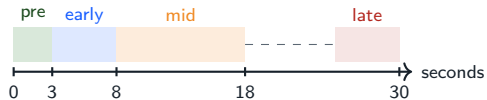
$$t_0 = \min(t), \quad t_1 = \max(t), \quad t_p = t_0 + 3$$

$$\mathcal{T}_{pre} = \{t : t \leq t_p\}$$

$$\mathcal{T}_{early} = \{t : t_p < t \leq t_p + 5\}$$

$$\mathcal{T}_{mid} = \{t : t_p + 5 < t \leq t_p + 15\}$$

$$\mathcal{T}_{late} = \{t : t \geq \max(t_p, t_1 - 5)\}$$



Temporal alignment

Event labels are window labels, but electrical signatures are time-shaped. These masks let us compare the same signal before and after the disturbance.

Baseline and robust scale are computed locally



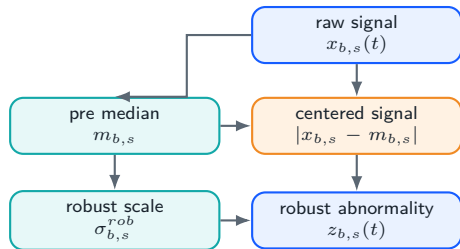
For each bus b and signal s

$$m_{b,s} = \text{median}_{t \in \mathcal{T}_{pre}} x_{b,s}(t)$$

$$\tilde{x}_{b,s}(t) = x_{b,s}(t) - m_{b,s}$$

$$\sigma_{b,s}^{rob} = 1.4826 \cdot \text{median}_{t \in \mathcal{T}_{pre}} |x_{b,s}(t) - m_{b,s}|$$

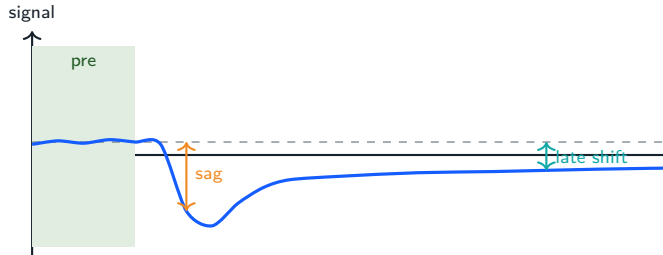
$$z_{b,s}(t) = \frac{|x_{b,s}(t) - m_{b,s}|}{\max(\sigma_{b,s}^{rob}, \epsilon)}$$



Implementation detail

If the MAD is too small, the code falls back to a standard deviation scale. If that also fails, it uses a safe fallback value.

Baseline subtraction, visually



Model input

The feature extractor does not keep this curve. It records quantities such as early span, derivative, maximum deviation, and late shift.

Fault reading

A sharp voltage sag after the pre segment creates large derivative and large robust z-score features.

Canonical pattern

`<PMU/BUS>__<SIGNAL>__<SEGMENT>__<STATISTIC>`

Examples

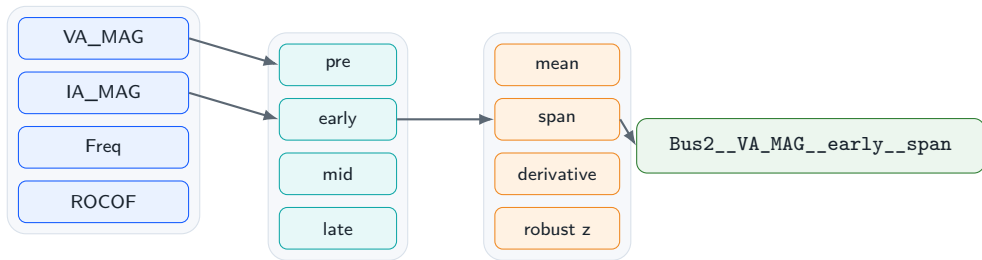
`Bus2__VA_MAG__early__span`
`Bus10__Freq__late__mean`
`GLOBAL__single_signal_score_max`

Reimplementation note

The feature vector has tens of thousands of columns. A consistent naming scheme makes debugging and reimplementation possible.

Family	Computed from	Physical signal
Segment statistics	every signal and segment	response magnitude and variability
Temporal deltas	early/mid/late vs pre	operating-point movement
Derivatives	finite differences	event abruptness
Robust z-score	pre-event median/MAD	abnormal signal magnitude
Data quality	gaps, NaNs, data-present	measurement availability
Single-signal anomaly	ranked abnormal channels	isolated channel corruption
Power proxy	V/I magnitudes and angles	active/reactive proxy movement
Global aggregations	PMU-level feature summaries	system-wide spatial pattern

Feature grid: signal × segment × statistic



Reimplementation shortcut

Write the extractor as nested loops: for each PMU, for each signal, for each segment, compute the same statistics with the same names.

For centered values $\tilde{x}(t)$ in segment $\mathcal{S}$

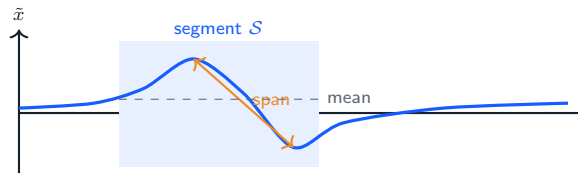
$$\mu_{\mathcal{S}} = \frac{1}{|\mathcal{S}|} \sum_{t \in \mathcal{S}} \tilde{x}(t)$$

$$\sigma_{\mathcal{S}} = \sqrt{\frac{1}{|\mathcal{S}|} \sum_{t \in \mathcal{S}} (\tilde{x}(t) - \mu_{\mathcal{S}})^2}$$

$$\text{span}_{\mathcal{S}} = \max_{t \in \mathcal{S}} \tilde{x}(t) - \min_{t \in \mathcal{S}} \tilde{x}(t)$$

Also stored

min, max, median, $\max |\tilde{x}|$, $\text{mean}|\tilde{x}|$, $p05$, $p95$



2. Temporal deltas



Definitions

$$\Delta_{\text{early}} = \mu_{\text{early}} - \mu_{\text{pre}}$$

$$\Delta_{\text{mid}} = \mu_{\text{mid}} - \mu_{\text{pre}}$$

$$\Delta_{\text{late}} = \mu_{\text{late}} - \mu_{\text{pre}}$$

Interpretation

- ▶ early delta: onset shock.
- ▶ mid delta: transient settling.
- ▶ late delta: persistent operating-point change.

Timing retained

Two windows can have the same full-window mean but different temporal behavior. The deltas preserve event timing.

3. Derivative features



Finite-difference approximation

$$d_x(t) = \frac{x(t) - x(t - \Delta t)}{\Delta t}$$

$$\text{max_abs_derivative} = \max_t |d_x(t)|, \quad \text{rms_derivative} = \sqrt{\frac{1}{T} \sum_t d_x(t)^2}$$

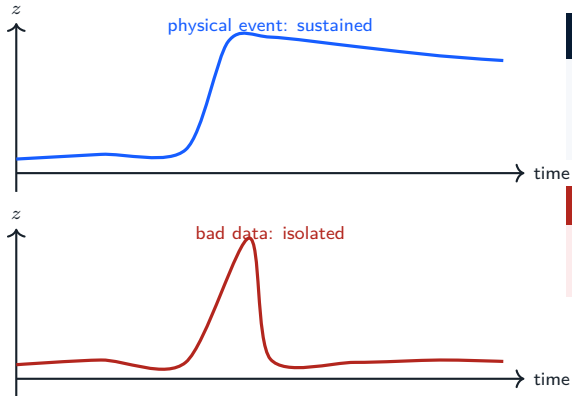
High derivative

Usually means an abrupt transient: fault onset, line switching, or corrupted data.

Event use

The final rule engine uses voltage and current derivative thresholds to guard fault and line-outage decisions.

Robust z-score separates two failure modes



Peak and persistence

`max_abs_robust_z` catches the spike. Sustained features say whether the abnormal value stays high for several samples.

Classifier use

The pair prevents isolated spikes from being treated as physical events.

5. Sustained anomaly features



Concept

Not every high z-score is equally meaningful. A single spike may be bad data; a sustained high z-score may be a physical event.

Short sustained score

`sustained_abs_robust_z_5`

Captures short bursts and sharp events.

Long sustained score

`sustained_abs_robust_z_15`

Captures persistent deviations after the event.

Anomaly interpretation

Faults can be sharp and sustained; bad data is often isolated to one channel; load/generation changes persist in late features.

6. Data-quality features



Features

- ▶ `n_samples, duration_s, dt_s`
- ▶ `max_timestamp_gap_s`
- ▶ `max_timestamp_gap_ratio`
- ▶ `data_present_fraction`
- ▶ `data_present_min`
- ▶ `missing_run_max_samples`
- ▶ `missing_run_max_s`
- ▶ `nan_fraction_max/mean`

Decision logic

missing $\Rightarrow \begin{cases} \text{Event 5,} & \text{no physical event} \\ \text{Event 6,} & \text{physical event also present} \end{cases}$

Physical meaning

Missing-data events are communication or measurement failures, not necessarily grid disturbances.

7. Single-signal bad-data features



Goal

Detect a channel that becomes extreme while the rest of the PMU/grid remains coherent.

Features

- ▶ `single_signal_score_max`
- ▶ `single_signal_score_second`
- ▶ `single_signal_score_ratio`
- ▶ `single_signal_score_count_gt20`

Rule of thumb

$$\frac{\text{top anomaly}}{\text{second anomaly}} \geq 3 \Rightarrow \text{bad data candidate}$$

8. Phase-spread and consistency features



Measured quantity

For voltage/current magnitude and angle groups, the extractor summarizes phase-to-phase spread and consistency.

Physical role

- ▶ three-phase imbalance,
- ▶ phase-angle inconsistency,
- ▶ PMU channel mismatch.

Event use

These features separate grid events from measurement artifacts because real events tend to preserve more cross-phase structure than corrupted channels.

Power proxy from voltage-current phase angle



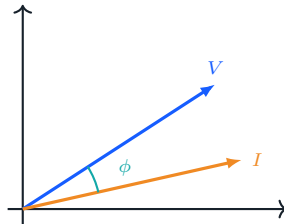
Per-unit magnitudes

$$V_{pu,\phi}(t) = \frac{|V_\phi(t)|}{\max(\text{median}_{pre}|V_\phi|, \epsilon)}$$

$$I_{pu,\phi}(t) = \frac{|I_\phi(t)|}{\max(\text{median}_{pre}|I_\phi|, \epsilon)}$$

Proxy, not calibrated power

The challenge data does not require a full power-flow solve. Sign depends on the PMU current convention, but the proxy gives the classifier a stable active/reactive signature.



$$\begin{aligned} P_{\text{proxy}} &= V_{pu} I_{pu} \cos \phi \\ Q_{\text{proxy}} &= V_{pu} I_{pu} \sin \phi \end{aligned}$$

Electrical reading

A load or generation change can shift I , ϕ , and frequency while voltage magnitude remains moderate.

10. Global PMU aggregation



Aggregated quantities

After extracting bus-level features, the model also computes global summaries across observed PMUs:

mean, max, min, std

System footprint

A local event has a spatial footprint. Global aggregation gives the classifier a system-level view: is the disturbance concentrated, widespread, coherent, or isolated?

Example

Bad data often has one extreme PMU/channel. A physical event affects multiple PMUs in a structured way.

Base feature count

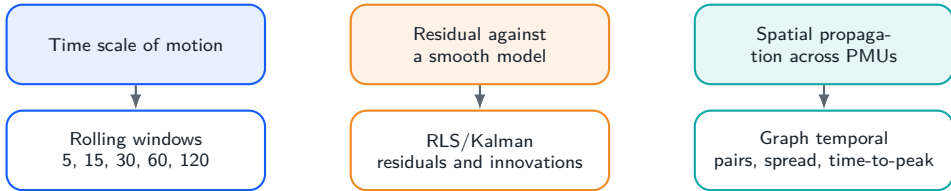


Feature group	Count
Voltage angle	7,596
Voltage magnitude	7,596
Current angle	7,596
Current magnitude	7,596
Frequency / ROCOF	4,640
Power proxy	4,428
Data quality	156
Single-signal anomaly	48
Other/global and alignment	88
Base v2 total	39,744

Reimplementation note

The base model uses a large but structured feature vector. The structure is what makes it reimplementable.

Dynamic feature blocks



Implementation

Each block is a separate feature module applied after base-window extraction.

Main signal set

$\{VA/VB/VC_MAG, IA/IB/IC_MAG, VA/VB/VC_ANG, IA/IB/IC_ANG, Freq, ROCOF\}$

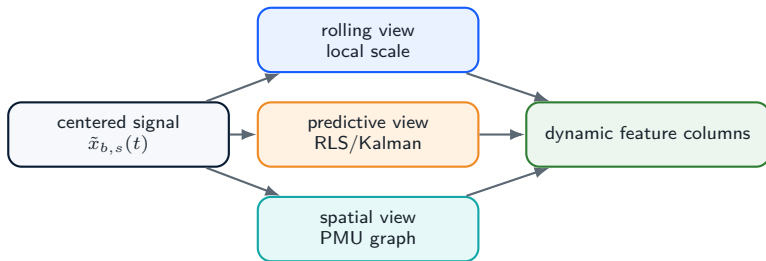
Reduced signal set for RLS/Kalman and graph features

$\{VA_MAG, IA_MAG, VA_ANG, IA_ANG, Freq, ROCOF\}$

Reason

The reduced set keeps dynamic features feasible while preserving representative voltage, current, angle, and frequency behavior.

Dynamic extractor: same signal, three views



Implementation reading

All dynamic blocks start after baseline subtraction, which keeps buses and RAW/SIM cases on comparable scales.

Rolling windows

$$w \in \{5, 15, 30, 60, 120\}$$

Features

- ▶ mean_abs_max
- ▶ std_max, std_mean
- ▶ slope_max_abs
- ▶ energy
- ▶ time_to_abs_peak

Physical interpretation

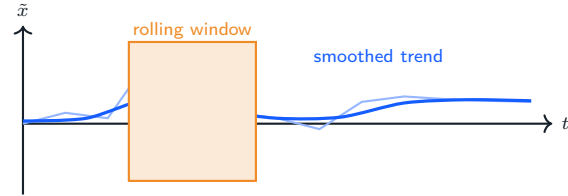
- ▶ short windows catch abrupt switching/fault onset;
- ▶ long windows catch sustained load/generation shifts;
- ▶ time-to-peak captures propagation delay.

Centered rolling mean and standard deviation

$$\mu_w(t) = \frac{1}{w} \sum_{\tau \in W_w(t)} \tilde{x}(\tau)$$

$$\sigma_w(t) = \sqrt{\frac{1}{w} \sum_{\tau \in W_w(t)} (\tilde{x}(\tau) - \mu_w(t))^2}$$

$$E_x = \frac{1}{T} \sum_t |\tilde{x}(t)|^2$$



Stored for each w

$$\max |\mu_w|, \quad \max \sigma_w, \quad \overline{\sigma_w}, \quad \max \left| \frac{d\mu_w}{dt} \right|, \quad E_x, \quad t_{\text{peak}}$$

Model

$$y_t = \theta_t + e_t$$

$$K_t = \frac{P_{t-1}}{\lambda + P_{t-1}}$$

$$\theta_t = \theta_{t-1} + K_t(y_t - \theta_{t-1})$$

$$P_t = \frac{P_{t-1} - K_t P_{t-1}}{\lambda}$$

Forgetting factors

$$\lambda \in \{0.92, 0.985\}$$

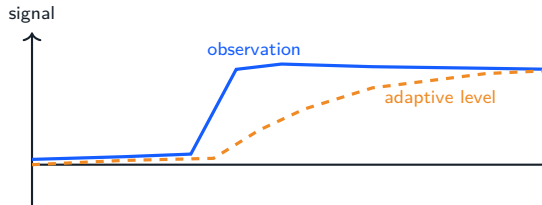
Features produced

$$r_t = |y_t - \theta_{t-1}|$$

$$\Delta\theta_t = |\theta_t - \theta_{t-1}|$$

For each sequence:

max, mean, p95, median_{late} - median_{pre}



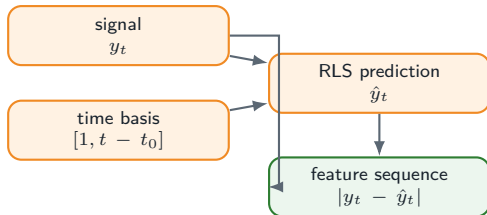
Local linear predictor

$$\phi_t = \begin{bmatrix} 1 \\ t - t_0 \end{bmatrix}, \quad \hat{y}_t = \phi_t^\top \theta_{t-1}$$

$$K_t = \frac{P_{t-1} \phi_t}{\lambda + \phi_t^\top P_{t-1} \phi_t}$$

$$\theta_t = \theta_{t-1} + K_t (y_t - \hat{y}_t)$$

$$P_t = \frac{P_{t-1} - K_t \phi_t^\top P_{t-1}}{\lambda}$$



Linear residual signal

They discount smooth ramps but react to switching, faults, and sudden sensor errors.

State-space view with assumed noise scales

$$x_t = x_{t-1} + w_t, \quad w_t \sim \mathcal{N}(0, Q)$$

$$y_t = x_t + v_t, \quad v_t \sim \mathcal{N}(0, R)$$

$$Q = (0.005 \text{ s})^2, \quad R = (0.08 \text{ s})^2$$

$$s = \text{robust_scale}(y)$$

Update equations

$$P_t^- = P_{t-1} + Q$$

$$\nu_t = y_t - \hat{x}_{t-1}$$

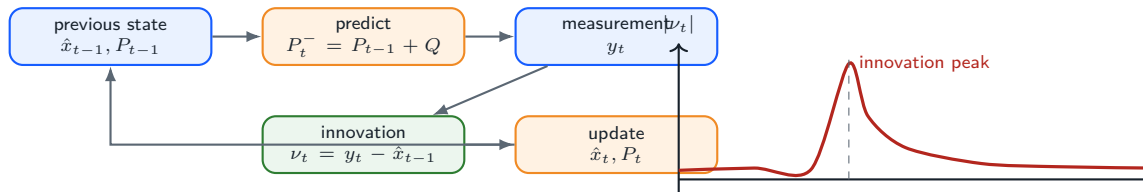
$$K_t = \frac{P_t^-}{P_t^- + R}$$

$$\hat{x}_t = \hat{x}_{t-1} + K_t \nu_t, \quad P_t = (1 - K_t) P_t^-$$

Feature sequence

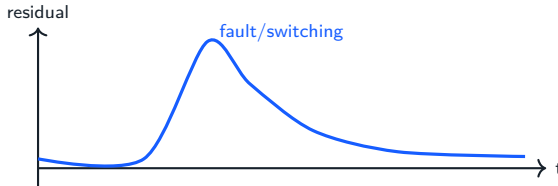
The stored signal is $|\nu_t|$, the absolute innovation. A large innovation means the measurement did not match the smooth level model.

Kalman innovation diagram



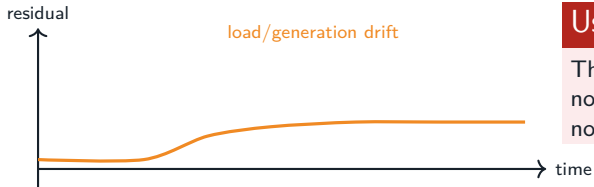
Stored features

$\max |\nu_t|, \quad \text{mean} |\nu_t|, \quad p95 |\nu_t|, \quad t_{\text{peak}}$



Residual pattern

Faults and switching tend to create a sharp innovation peak. Load and generation changes can create smaller but persistent residual deltas.



Used by final variant

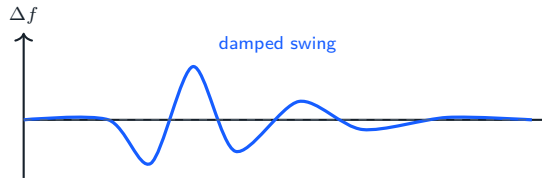
The selected localizer includes `rls_kalman`; the non-selected `swing_ekf` branch was tested but not used in the final route.

Classical generator intuition

$$\dot{\delta} = \omega - \omega_0$$
$$M\dot{\omega} = P_m - P_e - D(\omega - \omega_0)$$

Branch objective

Use a light extended-Kalman style dynamic residual to capture frequency and angle swings after generation/load disturbances.



Selection result

It improved some synthetic validation behavior but reduced RAW0001 localization. The final route kept the simpler RLS/Kalman innovation block.

For signal s and observed buses $\mathcal{B}$

$$\text{spread}_s(t) = \max_{b \in \mathcal{B}} \tilde{x}_{b,s}(t) - \min_{b \in \mathcal{B}} \tilde{x}_{b,s}(t)$$

$$\Delta_{ab,s}^{max} = \max_t |\tilde{x}_{a,s}(t) - \tilde{x}_{b,s}(t)|$$

$$\rho_{ab,s} = \text{corr}(\tilde{x}_{a,s}, \tilde{x}_{b,s})$$

$$\Delta t_{ab,s}^{peak} = |t_{a,s}^{peak} - t_{b,s}^{peak}|$$

Electrical weighting

$$w_{ab} = \frac{1}{\max(d_{ab}^{eff}, 10^{-5})}$$

$$G_s^{max} = \max_{a < b} w_{ab} \Delta_{ab,s}^{max}$$

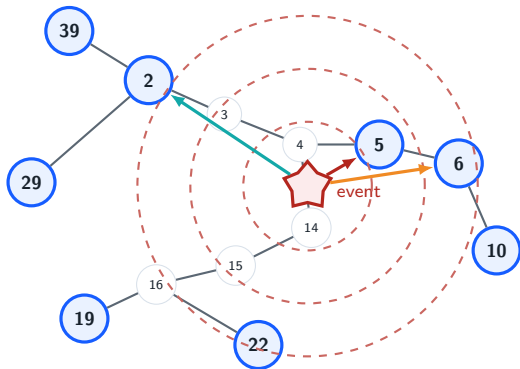
Feature names

GRAPH__VA_MAG__pmu_spread_max

GRAPH__Freq__pair_2_39__corr

GRAPH__IA_ANG__weighted_graph_diff_mean

Graph-temporal localization illustration



Nearby PMU signature

Larger amplitude, earlier peak, and stronger correlation with electrically close PMUs.

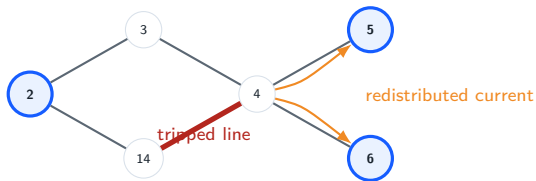
Remote PMU signature

Smaller response, delayed peak, or lower pairwise coherence.

Line outage reading

For Event 2, the localizer learns which line candidates match the current redistribution and graph-difference pattern.

Line outages require graph features



Electrical effect

Opening one line changes current paths. Voltage may not collapse like a fault, but current magnitude, current angle, and pairwise PMU differences move in a structured way.

Feature consequence

Event 2 benefits from graph differences, time-to-peak differences, and current-angle/current-magnitude signatures more than from a single largest voltage sag.

Selected

base_v2 + rolling + rls_kalman
+ graph_temporal

$$n_{\text{features}} = 45,162$$

Not selected

- ▶ algebraic_residual_v4: matched RAW0001 but did not improve it.
- ▶ swing_ekf: improved SIM slightly, reduced RAW0001 localization.
- ▶ oracle topology ranker: higher RAW0001, but not deployable.

Selection criterion

Tied for top deployable RAW0001 localization while keeping SIM validation performance in the selected tier.

ExtraTrees decision rule

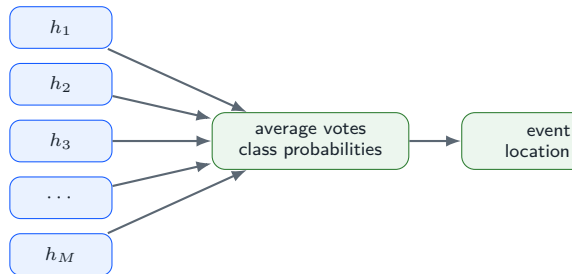
Ensemble probability

$$\hat{p}(c | x) = \frac{1}{M} \sum_{m=1}^M \mathbb{I}[h_m(x) = c]$$

$$\hat{c} = \arg \max_c \hat{p}(c | x)$$

ExtraTrees split policy

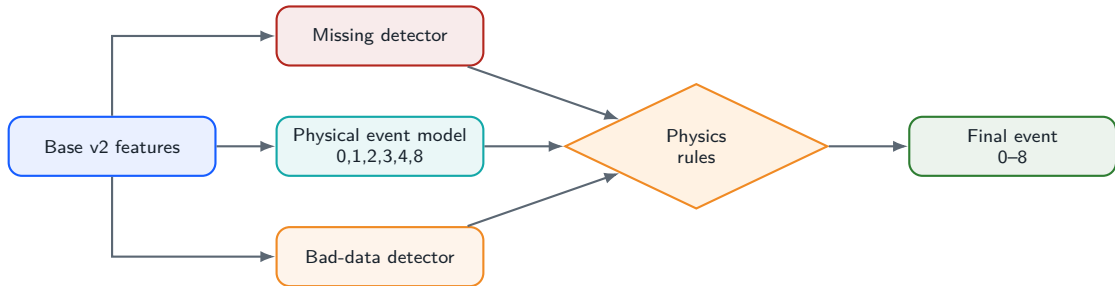
Each tree uses randomized split candidates. That reduces variance and works well when many engineered features are redundant but useful.



Pipeline detail

Before the forest, missing values are replaced by the training median of each feature column.

Model artifact	Role	Output
physical_event_classifier.joblib	Main physical event classifier	0,1,2,3,4,8
bad_data_detector.joblib	Event 7 detector	bad / not bad
missing_composition_detector.joblib	Event 5/6 support	missing composition
final_dynamic_localizers.joblib	Typed localizers	BUS, LINE, PMU
hierarchical_model_config.json	Rule thresholds	final event logic



Hierarchical routing rationale

Missing data and bad data are measurement phenomena. They should not be forced into the same flat classifier decision as physical grid events.

Fault-like

$$V_{\text{span}} \geq 50,000 \quad \text{or} \quad \max |dV/dt| \geq 500,000$$

Line-outage-like

$$\begin{aligned} I_{\text{span}} &\geq 500 \\ \max |dI/dt| &\geq 3,000 \\ V_{\text{span}} &\leq 50,000 \end{aligned}$$

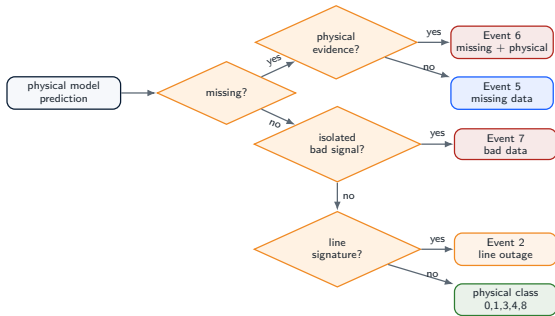
Missing-like

$\text{data_present_fraction} < 0.999$
or high NaN/gap features.

Bad-data-like

$$\begin{aligned} \text{single_signal_score} &\geq 200,000 \\ \text{single_signal_ratio} &\geq 3.0 \end{aligned}$$

Rule engine as a decision tree



Additional implementation gates

The implementation also uses confidence gates so weak physical predictions do not override strong missing or bad-data evidence.

Main evidence

- ▶ Voltage magnitude span.
- ▶ Voltage derivative.
- ▶ Robust z-score in voltage channels.
- ▶ Early segment changes.
- ▶ Cross-PMU severity pattern.

Manual recipe

1. Build baseline from pre-event voltage.
2. Detect large voltage sag/transient.
3. Reject missing-only and bad-data-only cases.
4. Localize to BUS using spatial severity.

Main evidence

- ▶ Current magnitude span.
- ▶ Current derivative.
- ▶ Current-angle changes.
- ▶ Voltage span not as extreme as a fault.
- ▶ Graph-temporal pairwise PMU differences.

Manual recipe

1. Look for current redistribution.
2. Guard against fault-like voltage collapse.
3. Use graph features to score candidate lines.
4. Localize to LINE.

Main evidence

- ▶ One signal has extreme robust z-score.
- ▶ Top anomaly is much larger than second anomaly.
- ▶ Other PMUs/channels remain coherent.
- ▶ Missing features do not dominate.

Localizer

Event 7 localizes to PMU, not BUS or LINE. The object that failed is the measurement device/channel source.

Main evidence

Event 8 captures cases with physical evidence that does not cleanly fit fault, line outage, generation change, or load change.

Useful features

- ▶ Power proxy.
- ▶ Rolling dynamics.
- ▶ RLS/Kalman residuals.
- ▶ Graph-temporal pattern.

Routing

The final runtime routes Event 8 to a BUS localizer because it represents a physical-grid signature rather than a PMU-only failure.

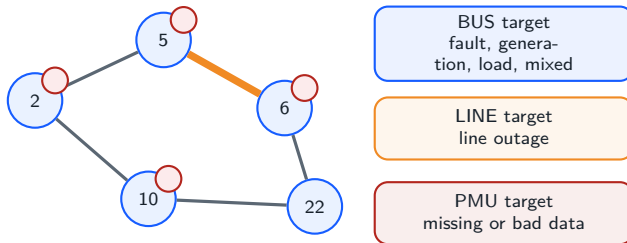
Event routing

Event	Localizer type
1 fault	BUS
2 line outage	LINE
3 generation change	BUS
4 load change	BUS
5 missing data	PMU
6 missing + physical	BUS or LINE
7 bad data	PMU
8 mixed proxy	BUS

Routing reason

A line outage, a bus fault, and a bad PMU are different localization problems. A single flat location classifier would mix incompatible targets.

Three location spaces



same PMU measurements, different target sets

Location semantics

The location column does not always mean a bus. Its meaning depends on the event class.

Localizer keys in the final dynamic bundle

BUS, BUS:event1, BUS:event3, BUS:event4, BUS:event8,
LINE, LINE:event2, PMU

Model family

Each localizer is a median-imputed ExtraTrees pipeline trained on base + dynamic features.

Final Top-1

`use_hybrid_top1 = false`. The final submitted Top-1 location uses the validated typed ML localizers, not the hybrid/oracle branch.

BUS	LINE	PMU
▶ nearest severe response, ▶ voltage/frequency signature, ▶ graph spread pattern.	▶ current redistribution, ▶ pairwise PMU differences, ▶ topology-aware graph signature.	▶ missing run, ▶ isolated bad channel, ▶ local data-quality failure.

Location signal

Location is inferred from the pattern of who saw the disturbance, how strongly, and how early.

Localizer inference equation

Feature vector for location

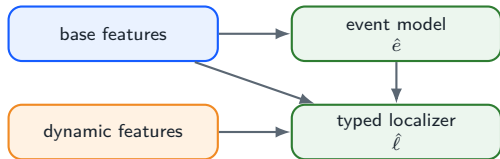
$$x_{\text{loc}} = [x_{\text{base_v2}}, x_{\text{rolling}}, x_{\text{rls/kalman}}, x_{\text{graph}}]$$

$$\hat{e} = H(x_{\text{base_v2}})$$

$$\hat{\ell} = L_{g(\hat{e})}(x_{\text{loc}})$$

Route function

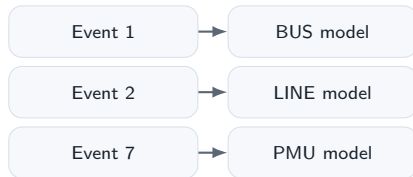
$$g(e) \in \{\text{BUS}, \text{LINE}, \text{PMU}\}$$



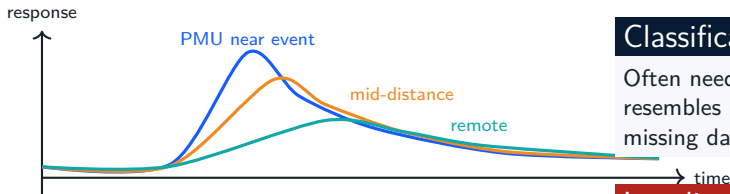
Reason for two feature views

The event classifier is stable with base features. The localizer gets the larger dynamic vector because location depends on propagation and topology.

BUS	LINE	PMU
$\mathcal{Y}_{BUS} = \{1, \dots, 39\}$ Target is the electrical bus where the physical disturbance is assigned.	$\mathcal{Y}_{LINE} = \{(i, j) : (i, j) \in \mathcal{E}\}$ Target is a branch in the IEEE 39 topology.	$\mathcal{Y}_{PMU} = \{2, 5, 6, 10, 19, 22, 29, 39\}$ Target is the observed device affected by missing or bad data.



Localization error sources



Classification

Often needs to know whether the pattern resembles a fault, outage, load change, missing data, or bad data.

Localization

Needs the finer spatial ordering: which PMU changed first, which changed most, and which candidate location best explains that pattern.

Training sources

- ▶ Synthetic SGSMA scenarios.
- ▶ Base features: `factory_features_v2.csv`.
- ▶ Dynamic features:
`dynamic_features_v3.csv`.
- ▶ Random state: 20260503.

Validation

- ▶ 30% stratified SIM holdout.
- ▶ RAW0001 transfer validation.
- ▶ Metrics: detector, classifier, macro-F1, localization.

Transfer principle

RAW labels were used for validation and analysis, not to train the final production-feasible submission path.

SIM holdout

Metric	Score
Detector accuracy	0.9693
Classifier accuracy	0.9673
Classifier macro-F1	0.9598
Localizer Top-1	0.8524

RAW0001

Metric	Score
Detector accuracy	1.0000
Classifier accuracy	1.0000
Classifier macro-F1	1.0000
Localizer Top-1	0.6667

Interpretation

Detection and classification transferred perfectly on RAW0001; localization remained the hard part.

Selected variant

`base_v2 + rolling + rls_kalman + graph_temporal`

Selection evidence

- ▶ Strong SIM performance.
- ▶ Tied best RAW0001 localization tier among deployable variants.
- ▶ No oracle RAW label use.
- ▶ Compact enough for submission.

Tradeoff

Some experimental methods had higher RAW0001 numbers, but were not valid for unseen data because they used RAW true labels or oracle assumptions.

DeepSeek role

The DeepSeek branch tested RAW0001 behavior and candidate localization ideas, but it was not the submitted inference route.

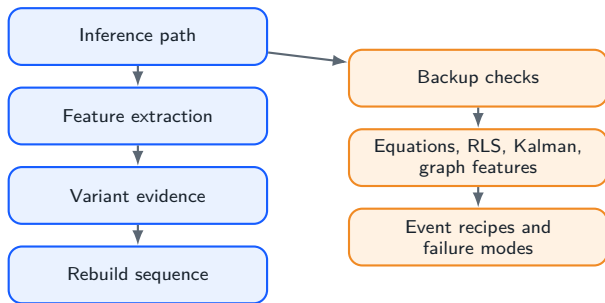
Excluded from inference

The best RAW0001 numbers from some DeepSeek variants used RAW labels directly for training, oversampling, weighting, or oracle ranking. That is not a valid inference setting for a new dataset.

Submitted method

The final submitted method was SGMS: physics-informed features + hierarchical ExtraTrees + typed dynamic localizers.

Evidence-to-implementation map

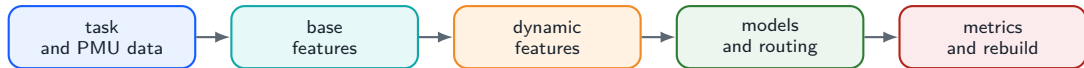


Primary path

Input windows, feature construction, event routing, and typed localization form the deployable method.

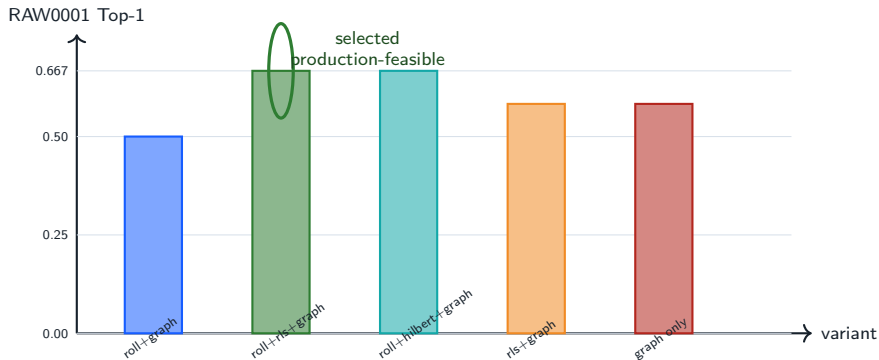
Backup path

Equations, event recipes, thresholds, and validation commands support implementation checks.



Runtime invariant

We do not classify raw waveforms directly. We turn PMU windows into physics-aware features, classify the event hierarchically, then localize it in the right target space.



Reading

The selected model kept perfect RAW0001 detection/classification and tied the top deployable RAW0001 localization tier without using RAW labels as training signal.

Feature mapping: robust abnormality



Equation

$$m = \text{median}(x_{pre})$$
$$s = 1.4826 \text{ median}|x_{pre} - m|$$
$$z(t) = \frac{|x(t) - m|}{\max(s, \epsilon)}$$

Code shape

```
m = median(pre)
s = 1.4826 * mad(pre)
z = abs((x-m)/max(s, eps))
```

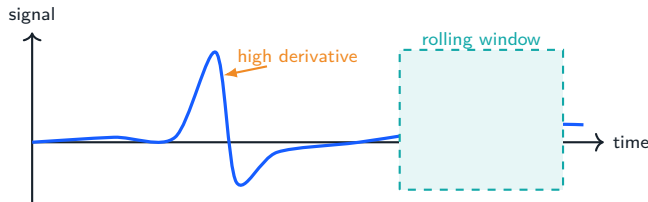
Feature names

```
max_abs_robust_z
sustained_abs_robust_z_5
sustained_abs_robust_z_15
```



Derivative

$$d_x(t_i) = \frac{\tilde{x}(t_i) - \tilde{x}(t_{i-1})}{\Delta t}$$
$$\max |d_x|, \quad \sqrt{\frac{1}{T} \sum_i d_x(t_i)^2}$$



Feature names

max_abs_derivative
rms_derivative
ROLL__w15__slope_max_abs

Code shape

```
dx = gradient(centered, dt)
roll = rolling(centered, w).mean()
roll_slope = gradient(roll, dt)
```

RLS residual

$$\hat{y}_t = \phi_t^\top \theta_{t-1}, \quad r_t = |y_t - \hat{y}_t|$$
$$\theta_t = \theta_{t-1} + K_t(y_t - \hat{y}_t)$$

Feature names

RLSK_*level_resid__max
RLSK_*linear_resid__p95
RLSK_*theta_delta__late_pre_delta

Kalman innovation

$$\nu_t = y_t - \hat{x}_{t-1}$$
$$K_t = \frac{P_t^-}{P_t^- + R}, \quad \hat{x}_t = \hat{x}_{t-1} + K_t \nu_t$$

Feature names

RLSK_*kalman_innov__max
RLSK_*kalman_innov__mean
RLSK_*kalman_innov__time_to_peak

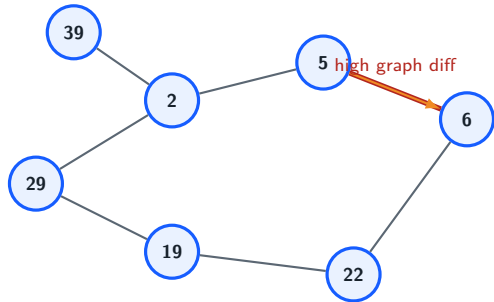


Selected pair relation

$$\Delta_{ab,s} = \max_t |\tilde{x}_{a,s}(t) - \tilde{x}_{b,s}(t)|$$

$$w_{ab} = \frac{1}{\max(d_{ab}^{eff}, 10^{-5})}$$

$$G_{ab,s} = w_{ab} \Delta_{ab,s}$$



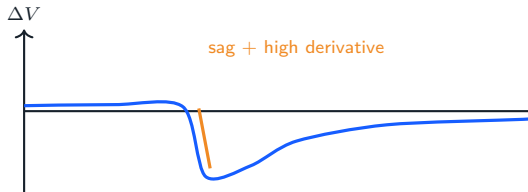
Feature names

GRAPH_*pmu_spread_max

GRAPH_*pair_a_b_corr

GRAPH_*weighted_graph_diff_mean

Event 1 recipe: fault



Minimum detector

$$\max |z_V| \uparrow, \quad \max \left| \frac{dV}{dt} \right| \uparrow$$

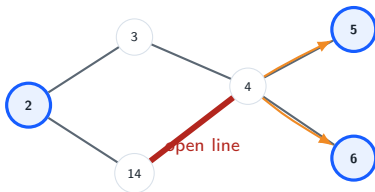
Use these features

VA/VB/VC_MAG__early__span
max_abs_derivative
sustained_abs_robust_z_5
GRAPH__*time_to_peak_*

Location

Route to BUS localizer. The bus target is inferred from severe nearby PMU response and graph timing.

Event 2 recipe: line outage



Minimum detector

Look for current redistribution and angle/current changes without the fault-level voltage sag.

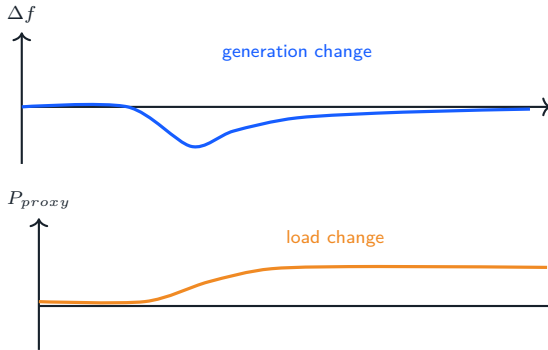
Use these features

IA/IB/IC_MAG__*
IA/IB/IC_ANG__*
P/Q_PROXY__*
GRAPH__*weighted_graph_diff_*

Location

Route to LINE localizer. The target is a branch (i, j) , not a bus.

Events 3 and 4 recipes: generation and load



Event 3

Generation change: frequency/ROCOF and power proxy shift. Route to BUS localizer.

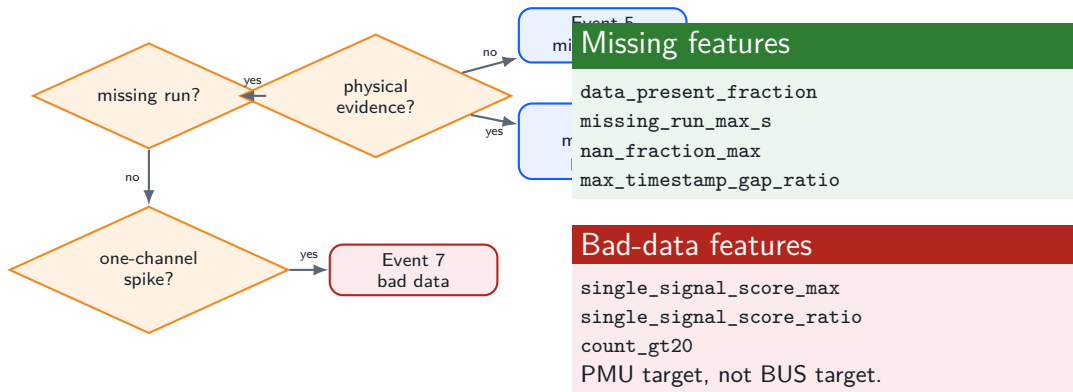
Event 4

Load change: sustained current and power proxy level shift. Route to BUS localizer.

Use these features

```
Freq__late__mean  
ROCOF__early__span  
P_PROXY__*late_pre_delta  
RLSK__*linear_resid_*
```

Events 5, 6, and 7 recipes: data failures



Event 8 recipe: mixed physical proxy

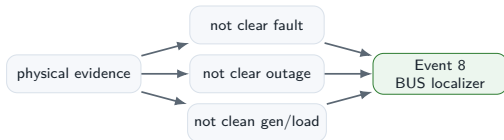


When it appears

The window has physical evidence, but it does not cleanly match fault, line outage, generation change, or load change.

Feature signature

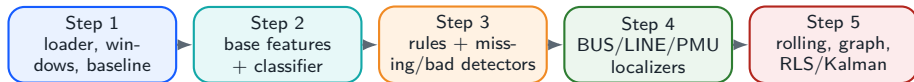
power proxy shifts
frequency/ROCOF evidence
graph spread
RLS/Kalman innovation



Location

Route to BUS. It is treated as a physical-grid event, not a PMU-only failure.

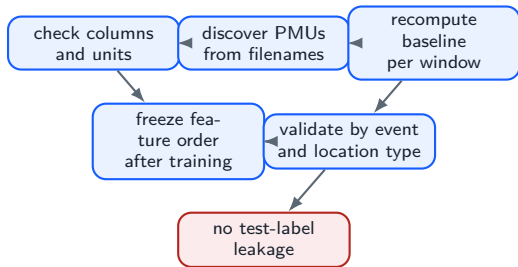
Implementation sequence for a new dataset



Ordering constraint

Build a working baseline early. Add dynamic features only after the event classifier and typed localizers produce valid CSV output.

Adapting to a new RAW dataset



Check first

Sampling rate, timestamp continuity, PMU list, channel names, angle units, and missing-data coding.

Keep stable

The physical feature definitions should stay the same. Retrain models and recalibrate thresholds if the new data distribution shifts.

```
case = read_pmu_case(input_dir)
windows = split_into_windows(case, seconds=30)

for window in windows:
    x_base = extract_window_features_v2(window)
    x_dyn = extract_dynamic_features(
        window,
        include_blocks=("rolling", "rls_kalman", "graph_temporal"),
    )

    physical = physical_model.predict(x_base)
    bad_score = bad_data_model.predict_proba(x_base)
    missing_score = missing_model.predict_proba(x_base)

    event = apply_physics_rule_engine(
        physical, bad_score, missing_score, x_base
    )

    loc_type = route_event_to_localizer(event, physical)
    location = localizers[loc_type].predict(concat(x_base, x_dyn))

    emit_rows_for_all_timestamps(window, event, location)
```

Build order from zero



1. Load all `Bus*.csv` files and align timestamps.
2. Create fixed 30-second windows.
3. Compute pre-event medians and unwrap angles.
4. Extract segment statistics, temporal deltas, and derivatives.
5. Add robust z-score and sustained anomaly features.
6. Add data-quality and single-signal bad-data features.
7. Add power proxy features.
8. Train physical ExtraTrees classifier.
9. Train missing and bad-data detectors.
10. Add physics rule guards.
11. Train BUS, LINE, and PMU localizers.
12. Add rolling and graph-temporal features.
13. Add RLS/Kalman if time remains.

Implement first

- ▶ 30 s windows.
- ▶ Pre-event baseline.
- ▶ Segment statistics.
- ▶ Derivatives.
- ▶ Robust z-score.
- ▶ Data-quality features.
- ▶ Power proxy.
- ▶ ExtraTrees + rules.

Then add

- ▶ Rolling dynamics.
- ▶ Graph-temporal PMU comparisons.
- ▶ RLS/Kalman residuals.
- ▶ Event-specific localizers.
- ▶ Top-K diagnostics.

Observed ablation pattern

Detection/classification can be strong with base features. Localization benefits the most from dynamic and graph-temporal features.

1. The decision unit is a 30-second PMU window.
2. The baseline is the pre-event median.
3. The features are physics-aware: V , I , angle, frequency, ROCOF, quality, and power proxy.
4. Dynamic features add temporal evolution and spatial propagation.
5. The classifier is hierarchical, not flat.
6. ExtraTrees is used because it handles many nonlinear, noisy features.
7. Localization is typed: BUS, LINE, or PMU depending on event.
8. RAW-label-tuned/oracle branches are excluded from the submitted inference route.

Submitted method

The final submission transforms raw PMU windows into physically meaningful feature vectors, then uses hierarchical ExtraTrees models and typed dynamic localizers to produce event and location predictions.

Feature engine

Base v2 + rolling +
RLS/Kalman + graph temporal.

Classifier

Physical event model + missing
detector + bad-data detector +
rules.

Localizer

Specialized BUS, LINE, and
PMU ExtraTrees localizers.

Selected features: `base_v2 + rolling + rls_kalman + graph_temporal`

Runtime bundle

- ▶ `models/final_model_config.json`
- ▶ `models/feature_columns.json`
- ▶ `models/classifiers/physical_event_classifier.joblib`
- ▶ `models/detector/bad_data_detector.joblib`
- ▶ `models/detector/missing_composition_detector.joblib`
- ▶ `models/detector/hierarchical_model_config.json`
- ▶ `models/localizer/final_dynamic_localizers.joblib`

Submission note

The large redundant hybrid localizer artifact was excluded from the compact zip; the final dynamic bundle is enough for the submitted Top-1 path.

Backup: exact physical thresholds



Parameter	Value
bad data probability	0.4211785821
missing composition probability	0.0
missing physical confidence	0.0
fault voltage span min	50,000
fault voltage derivative min	500,000
line current span min	500
line current derivative min	3,000
line voltage span max	50,000
bad single-signal score min	200,000
bad single-signal ratio min	3.0
line gate max physical confidence	0.75
fault guard max physical confidence	0.60

Final main classifiers

- ▶ Physical event classifier: ExtraTrees, 750 estimators, `max_features="sqrt"`, balanced class weights.
- ▶ Bad-data detector: ExtraTrees, 650 estimators, `max_features="sqrt"`, balanced class weights.
- ▶ Missing-composition detector: ExtraTrees, 650 estimators, `max_features="sqrt"`, balanced class weights.
- ▶ Dynamic localizers: ExtraTrees pipelines, 180 estimators in the final dynamic bundle.

Common preprocessing

`SimpleImputer(strategy="median")` before each ExtraTrees model.

Generic inference

```
python main.py -input-dir <RAW_OR_SIM_FOLDER> -output <OUTPUT_CSV>
```

SIM smoke test

```
python main.py -input-dir workbench\simulated\sgsma_generated\SIM00642\pmu -output  
workbench\_tmp\SIM00642_predictions.csv
```

RAW0001

```
python main.py -input-dir data\RAW0001 -output data\RAW0001\predictions.csv
```

Test suite

```
python -m pytest    Last reported local run: 246 passed, 10 skipped
```

Transfer mechanism

Physical feature transformations, simulated training scenarios, and hierarchical rules reduce dependence on one RAW folder. RAW labels remain evaluation evidence only.

Main risk

Localization is more sensitive than detection/classification because the spatial fingerprint depends on PMU placement, topology, and event location.

New-dataset calibration

Calibrate localizers and topology-aware guardrails using valid training labels from the new dataset, while avoiding test-label leakage.